Experiment No-6 : Vacuum Cleaner Problem

Aim:

To write a Python program to implement the Vacuum Cleaner problem in Artificial Intelligence.

Programming Code:

```
def vacuum_cleaner():
    rooms = {'A': 1, 'B': 1}
    location = 'A'

    while 0 in rooms.values():
        print("Vacuum is in room", location)

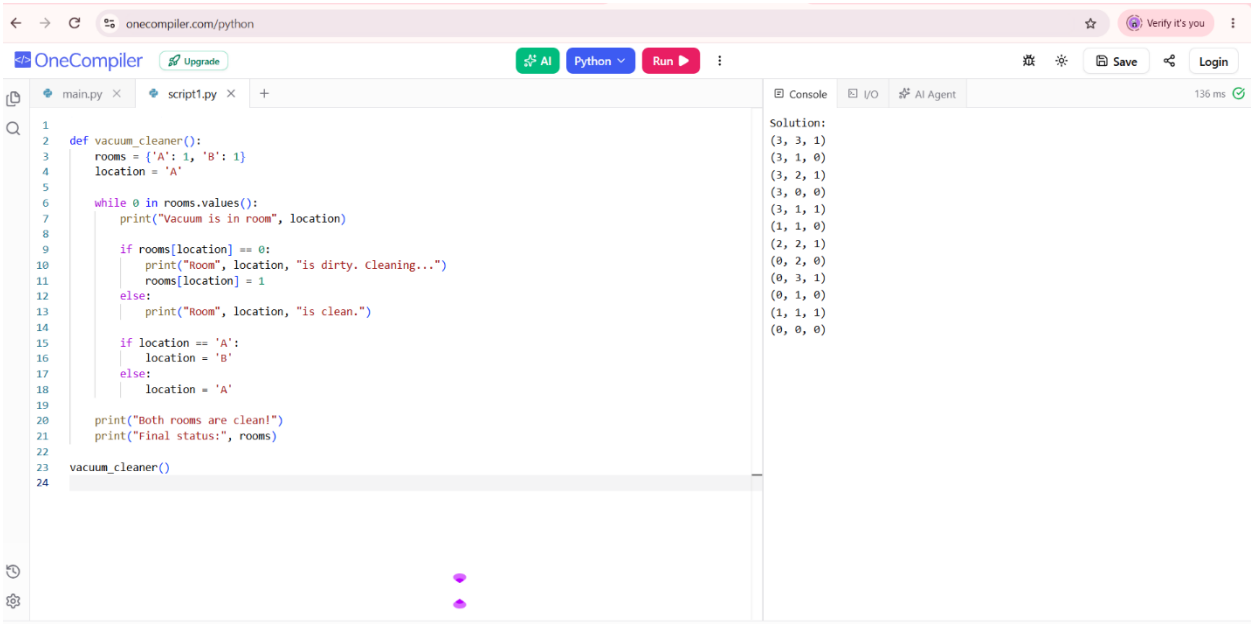
        if rooms[location] == 0:
            print("Room", location, "is dirty. Cleaning...")
            rooms[location] = 1
        else:
            print("Room", location, "is clean.")

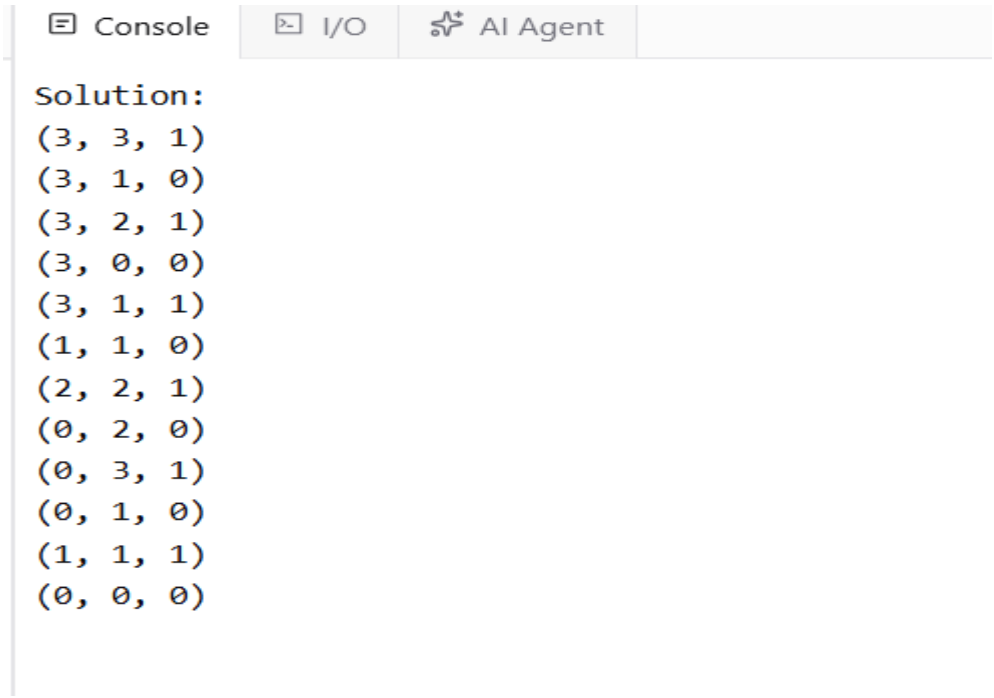
        if location == 'A':
            location = 'B'
        else:
            location = 'A'

    print("Both rooms are clean!")
    print("Final status:", rooms)

vacuum_cleaner()
```

OUTPUT:





Result

The Vacuum Cleaner problem was implemented successfully using Python. The program checks the rooms and cleans them until both rooms are clean.